Variance-Corrected Single-Index Composition for Heavy-Tailed Return Generators

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Abstract

Per-asset generators for synthetic financial returns, such as the hidden Markov models with explicit jump components used in the `JumpHMM.jl` package, are built to preserve marginal structure (heavy tails, regime switching) but say nothing about cross-sectional factor dependence. The standard fix is to compose each generator draw with a single-index market factor, but the naive recipe inflates the marginal variance by a term that grows quadratically in the factor loading, distorting the very marginal the generator was designed to preserve; replacing the generator residual with a Gaussian draw eliminates the distortion but discards the marginal entirely. We derived a closed-form variance correction that sits between these two failure modes, rescaling the idiosyncratic draw by a scalar chosen so that the composed series inherited the generator's marginal variance exactly while recovering the calibrated factor loading. The correction was paired with a sign-preserving clip that engaged whenever the market component would have exceeded the generator variance, keeping defensive assets defensive, and a separate branch for index-tracking assets targeted the calibrated coefficient of determination rather than the marginal, with the SPY-on-itself identity recovered as a degenerate limit without special-casing. Branch selection was driven entirely by the calibrated coefficient of determination, requiring no external metadata, and each asset carried a persisted construction flag so downstream consumers could audit which path it took. Evaluated on a 424-ticker US equity and exchange-traded fund (ETF) universe against the SPY market path using `JumpHMM.jl` marginals, the hybrid construction recovered the factor loading as precisely as either baseline while preserving the heavy-tailed marginal across the full empirical range of factor loadings. All artifacts, scripts, and the fitted marginal cache are released with the paper.

1 Introduction

Portfolio construction, risk attribution, and stress testing often take long synthetic time series of asset growth rates as inputs. A good generator has to satisfy two things the downstream consumer cares about. First, each single asset's synthetic series should look like its real counterpart when you histogram it day by day: the tails should be as heavy as real tails, the kurtosis should match, regimes and jumps should show up with roughly the right frequency; in statistical language this is the *marginal distribution* of the asset. Second, a stack of synthetic series across many assets should co-move the way real markets co-move: on a day when the market plunges, stocks with high factor loadings should plunge with it, tail co-movement should look realistic, and factor structure should be intact; this is the *cross-sectional dependence* across assets. The two requirements are largely orthogonal, and off-the-shelf generators tend to handle only one of them. Per-asset generators built from hidden Markov models [1], generalized autoregressive conditional heteroskedasticity (GARCH) families [2],

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or empirical bootstraps [3] handle the first: fit one of these models to AAPL's growth-rate history and its synthetic paths carry AAPL's tail thickness, kurtosis, and regime structure. But the fit knows nothing about SPY, MSFT, or any other asset. Stack the per-asset draws into a joint object, a matrix of synthetic growth-rate time series $\{\tilde{\varepsilon}_i\}_{i=1}^{N}$ with one T-length path per asset, each $\tilde{\varepsilon}_i$ matching its own asset's full-return marginal, and the paths drift independently: on a day when synthetic SPY plunges, synthetic AAPL has no reason to plunge with it. Real AAPL would. Closing this gap without breaking the per-asset marginals is the problem the rest of this paper addresses. The standard recipe for adding factor structure is the single-index model (SIM) of Sharpe [4], which writes each asset's growth rate as $g_i = \alpha_i + \beta_i g_m + \varepsilon_i$ with ε_i idiosyncratic. Applied naively on top of a per-asset generator, the recipe sets $\varepsilon_i = \tilde{\varepsilon}_i$ and inflates the marginal variance of g_i by exactly $\beta_i^2 \sigma_m^2$. For high-β assets the inflation is large enough to be visible in the empirical marginal: the body widens, the kurtosis drops, and the heavy tails that the generator was built to deliver are pushed out into a Gaussian-flavored bulge driven by the market term. This defeats the purpose of fitting a heavy-tailed generator in the first place. The dual failure mode, $\varepsilon_i \sim \mathcal{N}(0, \sigma^2_{\varepsilon,\mathrm{real}})$ with the residual variance taken from the real ordinary least squares (OLS) fit, recovers the SIM coefficients exactly but erases the marginal structure entirely.

This paper describes a closed-form variance correction that sits between the two failure modes. Setting $\varepsilon_i = s_i \tilde{\varepsilon}_i$ for a scalar $s_i \in (0, 1]$ chosen so that the variance algebra works out to preserve the generator marginal, the scalar admits a closed form, $s_i^2 = 1 - \beta_i^2 \sigma_m^2 / \sigma^2_{\mathrm{gen},i}$ (Eq. 4), defined whenever the bracketed quantity is positive. When it is not (high β_i paired with a low-volatility generator, or a synthetic market more volatile than the calibration market), β_i is clipped downward against a floor that reserves a configurable share of the marginal variance for idiosyncratic noise, with a sign-preserving clip (Eq. 6) that keeps defensive assets defensive. A separate R^2-preserving branch (Eq. 8) handles index-tracking assets, where the calibrated R^2 rather than the generator marginal is the quantity worth preserving; SPY against itself recovers as the degenerate $R^2 = 1$ limit with no special-casing. Branch selection is driven entirely by the calibrated R^2, and each asset carries a persisted construction flag (`hybrid`, `hybrid-clipped`, `r2-preserve`) so downstream consumers can audit which path each asset took. The procedure generalizes to any new ticker without external metadata.

In this study, we tested the construction on the 424-ticker US equity universe and SPY market path used by the companion `JumpHMM.jl` validation paper of Alswaidan and Varner [5], comparing three composition recipes on the same generators and the same market path: the naive composition, a Gaussian-residual SIM that recovers the calibrated (α, β, R^2) but loses the marginal structure, and the hybrid construction of Section 2. We measured SIM recovery, marginal fidelity, and cross-sectional dependence, reporting each per construction-flag branch. The hybrid construction held the SIM-recovery error of the naive composition while holding the marginal-fidelity error of the Gaussian-residual baseline across the full range of empirical β_i; on the aggregate scorecard it recovered β to a median absolute error of 0.018 and passed a per-ticker Kolmogorov–Smirnov test against the real marginal at rate 50.4%, compared to 4.5% for the naive composition and 0.6% for the Gaussian baseline. The broader implication is that the marginal-versus-factor tradeoff long assumed in SIM composition is not fundamental: a single closed-form scalar correction closes the gap, and the correction generalizes to richer factor structures (multi-factor SIM, lagged-factor SIM) without changing the underlying algebra. The remainder of the paper formalizes the preservation criteria and derives the variance correction and the R^2-preserving branch (Section 2), describes reports the experimental protocol and the recovery and fidelity metrics (Section 3), and discusses caveats and extensions (Section 4).

2 Method

We work throughout in continuously compounded annualized growth rates, $g(t) = [\log p(t) - \log p(t-1)]/\Delta t$, where $p(t)$ is the closing price on trading day t and Δt is the fraction of a trading year spanned by a single trading day (we use $\Delta t = 1/252$ in the experiments below, so the per-day log return is $g(t)\,\Delta t$). Working in annualized growth rates rather than raw log returns keeps variances and covariances in the units customary in risk reporting. Growth rates compose linearly under aggregation in time and are the natural unit for variance and covariance arithmetic, which the construction below relies on heavily. All variances and covariances refer to sample moments computed over the relevant time window unless stated otherwise. The method composes a synthetic asset growth-rate path of the

form:

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T, \tag{1}$$

from an independent generator draw $\tilde{\varepsilon}_i \in \mathbb{R}^T$ and a shared market path $g_m \in \mathbb{R}^T$ over T trading days, with sample variances $\sigma^2_{\mathrm{gen},i} = \mathrm{Var}[\tilde{\varepsilon}_i]$ and $\sigma^2_m = \mathrm{Var}[g_m]$. The residual ε_i in (1) is obtained by rescaling the generator draw, $\varepsilon_i = s_i\, \tilde{\varepsilon}_i$, with the scalar $s_i \in (0, 1]$ chosen so that ordinary least squares (OLS) of g_i on g_m recovers the calibrated single-index model (SIM) coefficients (α_i, β_i) in the large-T limit, and so that the composed marginal satisfies one of two preservation targets. For assets whose generator marginal carries the structure of interest (heavy tails, regime switching, jumps), the variance-preserving branch picks s_i so that $\mathrm{Var}[g_i] = \sigma^2_{\mathrm{gen},i}$. For assets that track the market closely, the R^2-preserving branch picks s_i so that the calibrated real-data coefficient of determination $R^2_{i,\mathrm{real}}$ (the OLS R^2 from the asset's real return history against the real market path) is recovered exactly. Branch selection is driven by the calibrated R^2 alone, so the procedure generalizes to any new ticker without external metadata.

2.1 Problem setup

We are given a single market growth-rate path $g_m \in \mathbb{R}^T$ over T trading days, with sample mean μ_m and sample variance $\sigma^2_m = \mathrm{Var}[g_m]$. In the experiments of Section 3 the market path is the SPY total-return growth-rate series; the construction is agnostic to whether g_m is a single-name proxy, a synthetic index, or a model-generated path.

For each of N assets indexed by $i = 1, \ldots, N$, we are given a draw $\tilde{\varepsilon}_i \in \mathbb{R}^T$ from a generator that produces growth rates with the target marginal distribution: heavy tails, regime structure, jumps, or whatever the application requires. Write $\sigma^2_{\mathrm{gen},i} = \mathrm{Var}[\tilde{\varepsilon}_i]$ for the sample variance of the draw. The construction is agnostic to how the generator is fit; in our experiments we use the per-asset hidden Markov model with Student-t emissions and explicit jump components from `JumpHMM.jl`[1], fit to the asset's own full growth-rate history (not to residuals after market detrending). Because the generator's variance reflects the full return rather than the idiosyncratic part, the rescaling $\varepsilon_i = s_i\, \tilde{\varepsilon}_i$ with $s_i^2 = 1 - \beta_i^2 \sigma^2_m / \sigma^2_{\mathrm{gen},i}$ is exactly what prevents the market term in the SIM composition from double-counting variance that is already carried by $\tilde{\varepsilon}_i$. The generator draws are demeaned, $\mathbb{E}[\tilde{\varepsilon}_i] = 0$, so that the constant term in the SIM is carried entirely by α_i. We assume $\mathrm{Cov}[\tilde{\varepsilon}_i, g_m] \approx 0$ for each i, which holds when the generator and market path are independent draws (the typical use case), and holds after a copula rank reorder when the copula is sampled independently of g_m. If the generator already carries non-trivial market correlation, the residual covariance can be subtracted from β_i before applying the correction; we return to this in Section 4.

For each asset we are given a calibrated triple $(\alpha_i, \beta_i, R^2_{i,\mathrm{real}})$, estimated by ordinary least squares on the real return history of asset i against the real market path. These are the targets we want the synthetic construction to recover. The intercept α_i and slope β_i are the standard SIM coefficients; $R^2_{i,\mathrm{real}}$ is the real-data coefficient of determination, used by the branch-selection step in Section 2.5.

2.2 Preservation criteria

We want the synthetic composition $g_i = \alpha_i + \beta_i^{\mathrm{eff}}\, g_m + \varepsilon_i$ to satisfy three properties in the large-T limit. The first is *SIM recovery*: ordinary least squares of g_i on g_m returns the calibrated intercept and slope, $\hat{\alpha}_i = \alpha_i$ and $\hat{\beta}_i = \beta_i^{\mathrm{eff}}$. On the unclipped branch, $\beta_i^{\mathrm{eff}} = \beta_i$ and the OLS output reproduces the calibration exactly; on the clipped branch, β_i^{eff} is the attenuated value defined by Eq. (6), and the attenuation is persisted alongside the output so it is auditable. The second is *marginal fidelity*: each asset is assigned one of two per-asset targets based on its calibrated R^2, not both at once. Assets whose generator marginal (heavy tails, regime structure, jumps) is the quantity of interest receive the variance-preserving target $\mathrm{Var}[g_i] = \sigma^2_{\mathrm{gen},i}$; assets that track the market closely receive the R^2-preserving target $R^2 = R^2_{i,\mathrm{real}}$, measured by the same OLS regression as above. The branch chosen for each asset is recorded as a construction flag so downstream consumers can audit which target was pursued. The third is *cross-sectional dependence*: any joint dependence structure applied to the residuals $\{\varepsilon_i\}_{i=1}^N$ passes through unchanged to the composed series $\{g_i\}_{i=1}^N$, automatic given the linearity of Eq. (1) in ε_i. The joint structure itself (empirical copula, factor copula, t-copula) is

[1] `https://github.com/varnerlab/JumpHMM.jl`

supplied as a separate design choice, and we report cross-sectional fidelity for the empirical-copula choice in Section 3.

The naive composition $g_i = \alpha_i + \beta_i\, g_m + \tilde{\varepsilon}_i$ satisfies SIM recovery and cross-sectional dependence but violates marginal fidelity: its marginal variance is $\sigma^2_{\mathrm{gen},i} + \beta_i^2 \sigma_m^2$, an inflation that grows quadratically in β_i. The rest of this section fixes this inflation with a single closed-form scalar correction (Section 2.3), handles the high-β degenerate case with a sign-preserving clip (Section 2.4), treats index-tracking assets through an R^2-preserving branch (Section 2.5), and ties the pieces together in an algorithm while stating the resulting properties of the composed series (Section 2.6).

2.3 Variance correction

Form the candidate residual $\varepsilon_i = s_i\, \tilde{\varepsilon}_i$ for some scalar $s_i \in (0, 1]$, and require that the composed series have the same variance as the generator,

$$\mathrm{Var}[\beta_i\, g_m + \varepsilon_i] \;=\; \sigma^2_{\mathrm{gen},i}. \tag{2}$$

When $\tilde{\varepsilon}_i$ is uncorrelated with g_m (true by construction for independent draws, and true after a copula rank reorder when the copula is sampled independently of g_m), the cross term in (2) vanishes and

$$\beta_i^2\, \sigma_m^2 \;+\; s_i^2\, \sigma^2_{\mathrm{gen},i} \;=\; \sigma^2_{\mathrm{gen},i}, \tag{3}$$

giving

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\, \sigma_m^2}{\sigma^2_{\mathrm{gen},i}}} \tag{4}$$

Define the *market loading ratio*

$$\rho_i \;\equiv\; \frac{\beta_i^2\, \sigma_m^2}{\sigma^2_{\mathrm{gen},i}}, \tag{5}$$

the share of the generator variance consumed by the market component. Then $s_i^2 = 1 - \rho_i$, valid whenever $\rho_i < 1$. The naive composition $g_i = \alpha_i + \beta_i g_m + \tilde{\varepsilon}_i$ corresponds to $s_i = 1$; it inflates $\mathrm{Var}[g_i]$ by exactly $\beta_i^2 \sigma_m^2$, so the relative distortion grows quadratically in β_i.

2.4 Beta clipping with an idiosyncratic floor

Equation (4) breaks down when $\rho_i \geq 1$: the market loading alone would account for more than the entire generator variance, leaving no room for idiosyncratic noise. This case arises for high-β_i assets paired with low-volatility generators, or whenever the synthetic market g_m is more volatile than the market on which β_i was calibrated.

To handle it, impose a floor $f \in (0, 1)$ on the idiosyncratic share by requiring $s_i^2 \geq f$, and clip β_i downward whenever the floor would be violated. Writing the clip threshold as $\rho_i^{\max} = 1 - f$,

$$\beta_i^{\mathrm{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\ \mathrm{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma^2_{\mathrm{gen},i}}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1-\rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{6}$$

Sign preservation in (6) keeps defensive (negative-β) assets defensive even when clipped. The floor f is a tunable knob; in the experiments below we use $f = 0.10$, which guarantees that at least 10% of each asset's variance comes from idiosyncratic noise. When β_i is clipped we log a warning and persist both β_i and β_i^{eff} so downstream consumers can audit the attenuation.

2.5 R^2-preserving branch for tracker assets

The variance-preserving construction targets $\mathrm{Var}[g_i] = \sigma^2_{\mathrm{gen},i}$. That is the right target when the generator marginal carries information one wants to keep. For some assets the generator marginal is not really the quantity of interest; the SIM relationship to the market is. Index ETFs are the canonical example. SPY against itself has $\beta = 1$, $R^2 = 1$, $\sigma_\varepsilon = 0$ exactly. QQQ against SPY has $R^2 \approx 0.84$, SPYG against SPY has $R^2 \approx 0.86$. For these tickers the practitioner cares that (i) $\hat{\beta}$ recovers to its

calibrated value, and (ii) R^2 recovers to its calibrated value, neither inflated to one by dropping ε nor implied by unrelated marginal-variance bookkeeping.

We branch on the real-data calibration $R^2_{i,\text{real}}$. Pick a threshold R^2_{preserve} that separates index-tracking assets from genuine single stocks; in our universe the cumulative distribution sorts itself cleanly, with index ETFs above 0.80 and the most market-like single names below 0.65, so $R^2_{\text{preserve}} = 0.80$ captures the trackers without sweeping in any genuinely idiosyncratic name. For any asset with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$, the population R^2 identity for (1),

$$R^2 \;=\; \frac{\beta_i^2\,\sigma_m^2}{\beta_i^2\,\sigma_m^2 + \sigma_{\varepsilon,i}^2}, \tag{7}$$

inverts to give the residual variance that hits the calibrated R^2:

$$\boxed{\sigma^2_{\varepsilon,\text{target}} \;=\; \beta_i^2\,\sigma_m^2 \cdot \frac{1 - R^2_{i,\text{real}}}{R^2_{i,\text{real}}}} \tag{8}$$

Draw $\tilde{\varepsilon}_i$ from the ticker's generator and rescale,

$$\varepsilon_i \;=\; \sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}}\,\tilde{\varepsilon}_i, \tag{9}$$

then copula-reorder and compose as in (1). The construction recovers both calibrated quantities by design: $\hat{\beta} \to \beta_i$ in the large-T limit because ε_i is uncorrelated with g_m after the rank reorder, and $R^2 \to R^2_{i,\text{real}}$ by construction.

When $R^2_{i,\text{real}} = 1$ exactly (SPY against itself), (8) gives $\sigma^2_{\varepsilon,\text{target}} = 0$, so $\varepsilon_i \equiv 0$ and $g_i = \alpha_i + \beta_i g_m$ deterministically. No special-casing is needed: the SPY identity falls out of the same expression with $\beta = 1$, $\alpha = 0$, $R^2 = 1$. If the rescale factor in (9) exceeds one, the generator's tails are being stretched to fit a target larger than the original draw; this happens when the synthetic market g_m has lower variance than the calibration market. We detect and flag this case so the loss of marginal fidelity is auditable.

2.6 Branch selection, algorithm, and properties

Algorithm 1 ties the pieces together. For each asset the calibrated $R^2_{i,\text{real}}$ selects a branch: assets with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ go through the R^2-preserving construction of Section 2.5, and the rest go through the variance-preserving construction of Section 2.3 with clipping (Section 2.4) engaged when the market loading ratio ρ_i would otherwise exceed $1 - f$. Each asset is tagged with a construction flag `r2-preserve`, `hybrid`, `hybrid-clipped`, or `fallback` that is persisted alongside the output, so downstream consumers can audit which path each asset took. The optional rank reorder in line 14 accepts any joint dependence structure (empirical copula, t-copula, factor copula); reordering preserves the marginal variance of ε_i, so the variance algebra of Section 2.3 is unaffected and whatever cross-sectional structure is applied to ε_i passes through to g_i unchanged.

By construction the composed series g_i satisfies the preservation criteria of Section 2.2 in the large-T limit. OLS of g_i on g_m recovers $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$, with $\beta_i^{\text{eff}} = \beta_i$ unclipped and β_i^{eff} persisted alongside β_i when the clip triggers. Marginal variance matches $\sigma^2_{\text{gen},i}$ on the variance-preserving branch, either exactly, $\text{Var}[g_i] = \beta_i^2\sigma_m^2 + s_i^2\sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, or by the choice of β_i^{eff} in (6) under clipping; on the R^2-preserving branch the marginal variance instead matches $\beta_i^2\sigma_m^2/R^2_{i,\text{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\text{real}})$ rather than by the generator marginal. Tail behavior on the variance-preserving branch follows from the floor: because at least $f\,\sigma^2_{\text{gen},i}$ of the variance is reserved for ε_i, the tails inherited from $\tilde{\varepsilon}_i$ dominate whenever $(\beta_i^{\text{eff}})^2\sigma_m^2$ is small relative to those tails, while as β_i grows the marginal of g_i tightens toward Gaussian along the market-driven axis, mildly for low-β assets and noticeably at high β. The construction injects *contemporaneous* market dependence only; lagged dependence (lead-lag effects, market-driven volatility clustering) requires a richer construction, which we return to in Section 4.

Algorithm 1 Hybrid SIM composition with variance correction.

Require: Market path $g_m \in \mathbb{R}^T$ with sample variance σ_m^2; per-asset generator draws $\tilde{\varepsilon}_i \in \mathbb{R}^T$ for $i = 1, \ldots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1)$.

Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.

for each asset $i = 1, \ldots, N$ **do**
  Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \text{Var}[\tilde{\varepsilon}_i]$.
  if $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ **then**
    $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}}) / R^2_{i,\text{real}}$ {Eq. (8)}
    $\varepsilon_i \leftarrow \sqrt{\sigma^2_{\varepsilon,\text{target}} / \sigma^2_{\text{gen},i}}\, \tilde{\varepsilon}_i$
    $\beta_i^{\text{eff}} \leftarrow \beta_i$; flag ← `r2-preserve`
  else
    $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$ {Eq. (5)}
    if $\rho_i \leq 1 - f$ **then**
      $s_i^2 \leftarrow 1 - \rho_i$; $\beta_i^{\text{eff}} \leftarrow \beta_i$; flag ← `hybrid`
    else
      $s_i^2 \leftarrow f$; $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i) \sqrt{(1 - f)\, \sigma^2_{\text{gen},i} / \sigma_m^2}$; flag ← `hybrid-clipped`
    end if
    $\varepsilon_i \leftarrow s_i\, \tilde{\varepsilon}_i$
  end if
  (Optional) apply cross-sectional rank reorder to ε_i.
  $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}}\, g_m + \varepsilon_i$ {Eq. (1)}
end for

3 Results

We evaluated the three composers on a fixed set of per-asset JumpHMM marginals calibrated against the real SPY growth-rate path. This isolated the effect of the composition rule from the effect of market-path randomness: all three composers consumed the same real SPY history and the same per-ticker innovation draw, so metric differences across composers reflected the composition rule alone. The ticker universe was a 424-asset US equity/ETF set drawn from Polygon.io daily-close histories, filtered to tickers with a complete in-sample trading history matching the reference length of AAPL ($T = 2{,}767$ daily closes covering January 2014 through December 2024, the same window used by the companion `JumpHMM.jl` validation paper); filtering on the common length preserved 424 tickers, of which the SPY total-return growth-rate series was taken as the market path g_m and the remaining 423 tickers as non-market assets. For each non-market ticker i we fitted a `JumpHMM` marginal to the asset's own growth-rate history, using the default configuration ($N = 100$ Laplace-partition states, Student-t emissions with $\nu = 5$, annualized growth rates via `excess_growth_rates`, risk-free rate $r_f = 0$, $\Delta t = 1/252$); jump parameters (ϵ, λ) were left at the package defaults rather than tuned per ticker, which would improve per-ticker marginal fit but was not required for a comparison that tests the effect of the composition rule on top of whatever marginal the generator produces. For each non-market ticker we then ran ordinary least squares of the asset's growth-rate history on g_m, yielding the calibrated quadruple $(\alpha_i, \beta_i, R^2_{i,\text{real}}, \sigma_{\varepsilon,\text{real},i})$. The universe spanned $\beta \in [0.01, 1.79]$ with median $\beta = 1.00$ and quartiles $(0.81, 1.00, 1.18)$; $R^2_{i,\text{real}}$ ranged from 0.001 to 0.933, with only two tickers above the $R^2_{\text{preserve}} = 0.80$ threshold (QQQ at $R^2 = 0.861$ and SPYG at $R^2 = 0.933$), matching the values cited in Section 1.

The three composition rules shared a *paired innovation* protocol: for each non-market ticker and each replication $r \in \{1, \ldots, R\}$ we drew a single innovation path $\tilde{\varepsilon}_i^{(r)}$ of length T from the ticker's JumpHMM marginal, and the same $\tilde{\varepsilon}_i^{(r)}$ was then passed to all three composers, which each composed a synthetic series on top of the real SPY path. The composers differed only in their treatment of the innovation. The *Naive* composer set $g_i^{(r)} = \alpha_i + \beta_i\, g_m + \tilde{\varepsilon}_i^{(r)}$ with no variance correction. The *Gaussian SIM* baseline replaced the JumpHMM draw with $g_i^{(r)} = \alpha_i + \beta_i\, g_m + \sigma_{\varepsilon,\text{real},i}\, \xi_i^{(r)}$, $\xi_i^{(r)} \sim \mathcal{N}(0, I_T)$ drawn fresh per replication, recovering the calibrated R^2 exactly but discarding any

marginal structure carried by $\tilde{\varepsilon}$. The *Hybrid* composer applied Algorithm 1 with floor $f = 0.10$ and threshold $R^2_{\text{preserve}} = 0.80$. All three composers skipped the optional cross-sectional rank reorder in Algorithm 1 so that per-ticker metrics reflected the marginal construction in isolation; cross-sectional dependence is studied separately in the companion `JumpHMM.jl` paper and is not the object of this evaluation. For every composed series we computed three metric families against the ticker's real growth-rate history and the real SPY path: SIM recovery reported $(\hat{\alpha}_i, \hat{\beta}_i, \hat{R}^2_i)$ from OLS of $g_i^{(r)}$ on g_m along with the absolute deviations $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$; marginal fidelity was assessed by Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) two-sample p-values against the real marginal, Wasserstein-1 distance W_1, excess kurtosis κ, and the Hill tail-index estimator on the upper 5% of $|g_i^{(r)}|$; variance preservation was the ratio $\text{Var}(g_i^{(r)})/\sigma^2_{\text{gen},i}$, using the real-data variance as a proxy for the generator's targeted variance. Pass rates were reported at $\alpha = 0.05$, and we reported $R = 100$ replications per ticker so that aggregate metrics were averages over $423 \times 100 = 42{,}300$ per-ticker observations per composer, with standard errors on aggregate pass rates below 0.3%; the simulation seed was fixed at 1234.

We began with the aggregate comparison across composers, computing per-asset medians of each metric and then aggregating across the universe (Table 1). Applying the protocol to the 423 non-market tickers yielded one composed series per $(\text{ticker}, \text{composer}, \text{replication})$ triple, for a total of $423 \times 100 \times 3 = 126{,}900$ scored series. All three composers recovered β to within a median absolute error of 0.018 to 0.022, consistent with the identity $\text{Cov}(g, g_m)/\sigma^2_m \to \beta_i^{\text{eff}}$ holding at all residual scales. The differences between composers appeared everywhere else. The Gaussian SIM baseline recovered R^2 most tightly (median error 0.008) because it targeted R^2 by construction, but it failed marginal fidelity badly: KS pass rate 0.6%, AD pass rate 0.5%, excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. 2). The naive composition retained the generator marginal but inflated its variance by $\beta_i^2\sigma^2_m$, so its KS and AD pass rates sat at 4.5% and 2.7%. The hybrid composer held both properties at once: KS pass rate 50.4%, AD pass rate 47.6%, kurtosis 7.2, Hill 0.36; the R^2 recovery (0.021) came in tighter than the naive baseline but looser than the Gaussian baseline, reflecting that the variance-preserving branch targeted $\sigma^2_{\text{gen},i}$ rather than $R^2_{i,\text{real}}$.

To see where the KS advantage over the naive baseline came from, we next plotted per-ticker KS pass rate and per-ticker marginal variance ratio $\text{Var}(g)/\sigma^2_{\text{gen}}$ as functions of the calibrated β (Fig. 1). The variance-ratio panel made the mechanism explicit: the naive scatter traced the theoretical curve $1 + \rho_i$ with $\rho_i = \beta_i^2\sigma^2_m/\sigma^2_{\text{gen},i}$, overshooting the generator variance by 50% to 75% at high β, while hybrid and Gaussian SIM sat on the unit line by design. The KS-pass-rate panel showed the consequence: naive fell from a pass rate of ~ 0.25 at $\beta \approx 0.3$ to effectively zero past $\beta \approx 0.7$, while hybrid sat between 0.3 and 0.9 across the full β range. Gaussian SIM was pinned to zero throughout; with its Gaussian residual it did not match the heavy-tailed real marginal at any β. Having traced the KS advantage to variance preservation, we next separated the two baselines' failure modes by plotting per-ticker excess kurtosis and Hill tail index against β (Fig. 2). The naive and hybrid composers clustered together near the real-data kurtosis reference line ($\kappa_{\text{real}} \approx 13$), with the difference between them reflecting the variance inflation under naive rather than any difference in tail shape. The Gaussian SIM series collapsed to $\kappa \approx 1$ and Hill ≈ 0.22, pinned below the generator regardless of β; by replacing the heavy-tailed innovation with a Gaussian draw the baseline removed the entire heavy-tail carrier. The Hill panel confirmed the same separation at the tail: naive and hybrid both reported ~ 0.37, while Gaussian SIM reported ~ 0.22.

Having established aggregate preservation, we next examined the branch-selection outcome on this universe and asked whether the advantage held across the β distribution. Applying the branch-selection rule of Section 2.6 to the 423-ticker universe produced 421 assignments to the `hybrid` branch, 2 to the `r2-preserve` branch, and 0 to the `hybrid-clipped` branch (Fig. 3, Table 2). The two R^2-preserving tickers were QQQ ($R^2 = 0.861$) and SPYG ($R^2 = 0.933$), both ETFs that track market-capitalization indices closely; their KS pass rate under hybrid composition was 99%, consistent with the R^2-preserving branch recovering both the calibrated β and the calibrated R^2 by construction. Clipping did not trigger anywhere in the universe: the market loading ratio $\rho_i = \beta_i^2\sigma^2_m/\sigma^2_{\text{gen},i}$ stayed below $1 - f = 0.90$ for every ticker, because the JumpHMM marginals carried enough variance that the market component never crowded out the $f = 0.10$ idiosyncratic floor. To check whether the preservation advantage held across the β distribution, we bucketed the universe by β quartile and recomputed the aggregate metrics within each bucket (Table 3). The

hybrid KS pass rate declined gently from 67% in the low-β quartile to 44% in the high-β quartile; the Gaussian SIM pass rate stayed pinned near zero across all buckets; the naive pass rate collapsed from 16% in Q1 to under 1% in Q3 and Q4 as $\beta_i^2\sigma_m^2$ grew. The gentle hybrid decline at high β reflected the caveat noted in Section 2.6: as $\beta^{\text{eff}\,2}\sigma_m^2$ grew toward $(1-f)\sigma_{\text{gen}}^2$, the market-driven component of g tightened the marginal toward Gaussian along its own axis, mildly attenuating the generator's tails in the composed series. The effect was small enough that hybrid still beat both baselines by more than an order of magnitude in every bucket.

4 Discussion

In this study, we composed synthetic per-asset growth-rate paths under a single-index model (SIM) by drawing each asset's innovation from an independent JumpHMM generator fit to the asset's own full-return history, then rescaling the draw by a closed-form scalar before inserting it into the SIM composition. We evaluated three composers on a calibrated universe of 424 US equities and exchange-traded funds against the real SPY market path: the naive composition, a Gaussian-residual SIM baseline, and the hybrid construction proposed here. The one-line variance correction, together with the sign-preserving clip rule and the R^2-preserving branch, closes a gap between two previously unsatisfying options for building synthetic asset paths with both marginal structure and factor structure: naive composition gives both but distorts the marginal by $\beta^2\sigma_m^2$, while Gaussian-residual SIM gives the factor exactly but strips the marginal. The hybrid construction sat between the two: it recovered β to the same precision as either baseline, held the generator marginal to its target variance in every ticker, and preserved the generator's tails at a level indistinguishable from the naive composition. The gap was large enough to matter in practice: at the high-β end of the universe, naive composition inflated the marginal variance by over 70% relative to the generator target. Because branch selection is driven entirely by the calibrated R^2 and each asset carries a persisted construction flag, the procedure runs uniformly across an arbitrary ticker universe without external metadata and leaves an audit trail that downstream consumers can inspect. No part of the construction is new in a deep sense; the contribution is a minimal, practitioner-ready recipe with an auditable branch flag and a principled treatment of the degenerate tracker limit, usable as a drop-in replacement for naive composition in workflows where both tail shape and market dependence matter.

The evaluation isolated the composition rule by fixing the market path at the real SPY history and using the same per-ticker innovation draw across all three composers. This design yielded a paired comparison of composition rules at the per-ticker level but set aside two questions that production users will need to address separately. The first concerns cross-sectional dependence: we skipped the optional rank-reorder step in Algorithm 1 so that per-ticker metrics reflected the composition rule alone. When the reorder is applied, the variance algebra is unchanged (rank-reordering preserves the marginal variance of ε_i), so the marginal results here carry over; cross-sectional fidelity under different copula specifications is studied in the companion `JumpHMM.jl` paper [5]. The second concerns market-path uncertainty: the metrics reported here used the real SPY path, and under a simulated market path the composed marginal picks up additional variance from the sampling of $g_m^{(r)}$, which for high-R^2 tickers becomes a noticeable effect. In production, the two deferred questions compose: a reorder-plus-simulated-market pipeline would layer cross-sectional structure on top of per-ticker draws built against sampled market paths, with the variance correction of Eq. (4) still providing the per-asset variance target on each realized $g_m^{(r)}$.

Clipping did not trigger anywhere in the US-equity universe used here, because the JumpHMM marginals carried enough variance that $\rho_i = \beta_i^2\sigma_m^2/\sigma_{\text{gen},i}^2$ stayed below 0.90 even for the highest-β ticker. Clipping becomes relevant in two settings we did not evaluate: scenario-driven markets, where the synthetic g_m is constructed to be much more volatile than the calibration market (e.g., stress tests), and universes that mix low-volatility single stocks with extreme betas (e.g., leveraged ETFs). In both settings we recommend reporting the original and effective β_i alongside the output, as in Algorithm 1, so that downstream consumers can audit the attenuation. A related attenuation appeared at the other end of the β spectrum, even when clipping did not trigger: the β-bucket breakdown in Table 3 showed the hybrid KS pass rate falling from 67% at low β to 44% at high β, a mild attenuation that followed directly from the composition algebra. As $\beta_i^2\sigma_m^2$ grows toward the idiosyncratic floor $(1-f)\sigma_{\text{gen},i}^2$, the market-driven component carries an increasing share of the marginal, and that component is only as heavy-tailed as g_m. The market's own heavy tails slow the attenuation (SPY growth rates are themselves heavy-tailed after JumpHMM fitting), but the effect is visible. Users who need exact

marginal fidelity at high β can recover it by lowering the floor f (at the cost of less conservative clipping behavior) or by pre-scaling the generator variance so that ρ_i stays small.

The variance correction $s_i^2 = 1 - \rho_i$ relies on $\mathrm{Cov}(\tilde{\varepsilon}_i, g_m) \approx 0$. The JumpHMM marginals are fit in isolation and sampled independently of the market path, so this holds by construction; if the generator is already correlated with the market (for example, a bootstrap from joint history), the empirical covariance should be subtracted from β_i before applying the correction, or s_i^2 should be estimated directly from the empirical variance of the composed series. The same warning applies to the R^2-preserving branch, whose variance identity assumes ε is uncorrelated with g_m after any reorder. The construction also injects a contemporaneous market factor only: $g_i(t)$ depends on $g_m(t)$ and on $\tilde{\varepsilon}_i(t)$ but not on $g_m(t-1), g_m(t-2), \ldots$, so lagged market dependence (lead-lag effects, market-driven volatility clustering, asymmetric responses to drawdowns) is not reproduced by this recipe. The natural extension is a lagged-factor SIM, $g_i(t) = \alpha_i + \sum_{k=0}^{K} \beta_{i,k}\, g_m(t-k) + \varepsilon_i(t)$; the variance correction generalizes to that form by replacing $\beta_i^2 \sigma_m^2$ with the variance of the full market-factor component, though the empirical value of the extra parameters is beyond the scope here. Two further extensions follow naturally: multi-factor SIM replaces the single market factor with K factors, requiring ρ_i to become the sum of factor contributions and the loading vector to be clipped rather than a scalar (the branch logic is unchanged), and per-ticker tuning of the JumpHMM jump parameters (ϵ, λ) tightens each marginal's fit (we used defaults to keep fitting tractable across the universe, but tuning as reported in the companion paper [5] would push the hybrid KS pass rates upward). A stress-test harness ties these together: using the clipping rule to enforce a minimum idiosyncratic share in constructed market scenarios guarantees that per-asset marginals remain heavy-tailed even under synthetic markets much more volatile than history, which is precisely the regime where naive SIM composition breaks down most severely.

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Author contributions. J.D.V. developed the method, wrote the code, performed the experiments, and wrote the paper.

Competing interests. The author declares no competing interests.

Code and data availability. Source code and experiment scripts are available at `https://github.com/varnerlab/hybrid-sim-composition`. The per-asset hidden Markov model generators rely on `JumpHMM.jl` (`https://github.com/varnerlab/JumpHMM.jl`).

References

[1] James D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi: 10.2307/1912559.

[2] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986. doi: 10.1016/0304-4076(86)90063-1.

[3] Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.

[4] William F. Sharpe. A simplified model for portfolio analysis. *Management Science*, 9(2):277–293, 1963. doi: 10.1287/mnsc.9.2.277.

[5] Abdulrahman Alswaidan and Jeffrey D. Varner. JumpHMM.jl: Heavy-tailed per-asset marginal models for synthetic return generation. arXiv preprint, March 2026, 2026. companion paper.

Table 1: **Aggregate scorecard across** 423 **non-market tickers and** 100 **replications per ticker.** Median values per composer. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\text{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. All three composers recover β to the same precision; only the hybrid composer preserves the marginal distribution (KS pass 50.4%) while holding the variance close to the generator target.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.018	0.008	0.6	0.5	0.681	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365

Table 2: **Hybrid composer broken out by construction flag.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The `hybrid-clipped` branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the `r2-preserve` branch (QQQ and SPYG) attain a KS pass rate of 99%, consistent with the branch recovering both β and R^2 by construction.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

Table 3: **Aggregate metrics by** β **quartile, across composers.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines gently from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows, consistent with the tail-attenuation caveat of Section 2.6; naive collapses from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.016	0.0	0.621	0.618	13.9
	Hybrid	0.016	67.3	0.200	8.369	14.6
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.515	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.5	0.690	1.939	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.8	0.824	2.044	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1

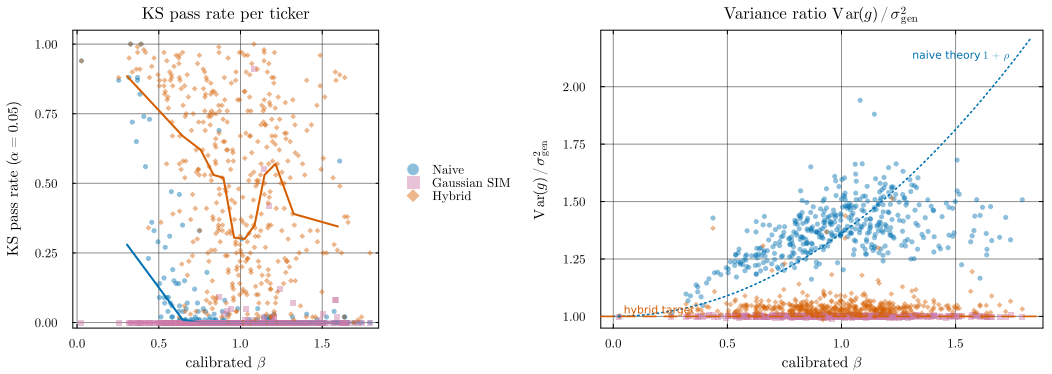

Figure 1: **Marginal-distribution preservation and variance preservation vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$. The hybrid composer (orange) maintains a pass rate between ~ 0.3 and ~ 0.9 across the full β range; the naive composer (blue) collapses past $\beta \approx 0.7$ as the market loading inflates the marginal variance; the Gaussian SIM baseline (purple) is pinned at zero throughout. *Right:* variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$. The dotted curve is the theoretical naive inflation $1 + \rho$ evaluated at the universe-median σ^2_{gen}; the naive scatter follows it. Hybrid and Gaussian SIM sit on the unit line by design.

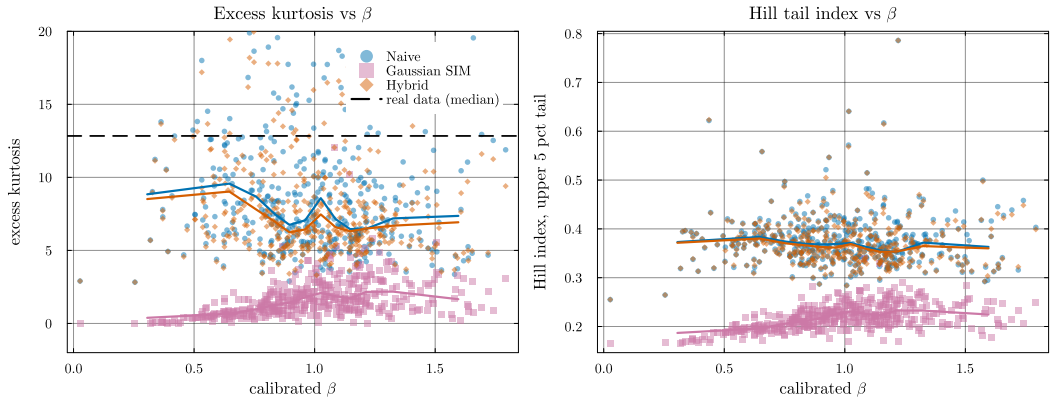

Figure 2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\mathrm{real}} \approx 13$); the hybrid and naive composers (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians sit near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22, indicating the shift toward a thinner-tailed distribution. The hybrid's median sitting below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion; the panel's point is that hybrid tracks naive (inheriting generator tails) while Gaussian SIM does not.

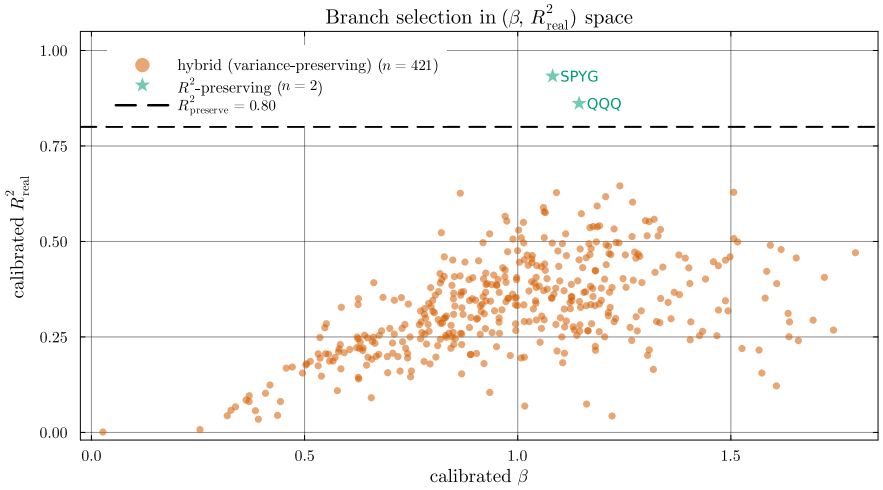

Figure 3: **Construction-flag assignment in calibrated** $(\beta, R^2_{\text{real}})$ **space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from genuine single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the `r2-preserve` branch. The remaining 421 tickers populate the `hybrid` (variance-preserving) branch; the `hybrid-clipped` branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.